The Link Function Problem in Psychological Interaction Testing

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Abstract

Equal differences on one scale become different differences on another scale. This is crucial for interaction testing because an interaction is a difference between differences, and in psychological research the observed response and the construct of interest are often expressed on different scales. Therefore, the same data can show an interaction on one scale and not on another. Generalized Linear Models make this problem explicit by separating two decisions, the family, which describes the nature of the response variable, and the link function, which sets the scale on which effects are additive. Because a model without an interaction term is additive on the link scale, choosing a link is choosing what 'no interaction' means. The paper has two connected parts. First, a preregistered review of recent articles from five prominent psychology journals shows that interaction testing is routinely performed in research, that the link function is most often unspecified, and that constrained outcomes are frequently analyzed with Gaussian identity-link models. Second, worked examples and simulations examine forced-choice accuracy with a chance floor, the choice between logit and probit links, and sum scores as a measurement case, and quantify how a plausible-looking link can turn an additive structure on one scale into a statistically significant interaction on another. A final diagnostic simulation shows that residual checks, a link test, and AIC comparisons give uneven protection and do not by themselves decide which scale the scientific claim concerns. Interaction claims remain underspecified until researchers state, justify, and probe the scale on which no interaction is assumed.

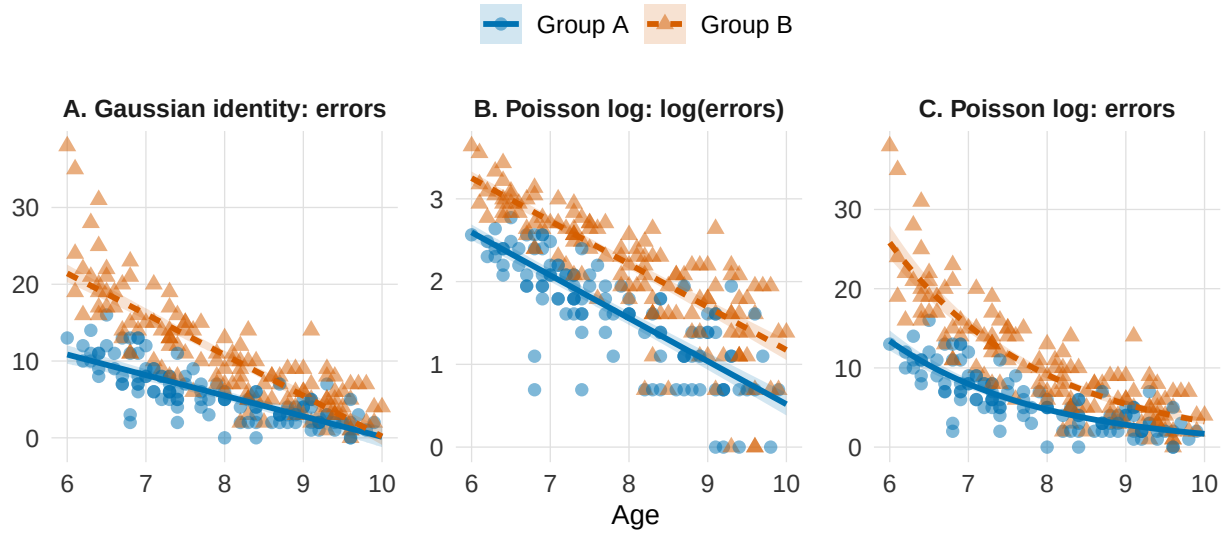
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The Link Function Problem in Psychological Interaction Testing

We start with a simplified example. Two groups of children, for instance a clinical and a neurotypical group, differ in how many errors they make on a task, and in both groups errors decrease with age. Imagine a researcher who records error counts from children aged six to ten and analyzes them, as is common, with a Gaussian linear model that includes an age-by-group interaction. In Figure 1 (Panel A) the two fitted lines decline at different rates, and the gap between the groups is much wider at age six than at age ten. Is there an age-by-group interaction?

Figure 1

Simulated error counts for two groups of children aged 6 to 10, generated with the same age effect in both groups on the log of the expected error count. A Gaussian linear model fitted to the counts produces straight lines with different slopes, which suggests an age-by-group interaction. A Poisson model with a log link fitted to the same data produces curves that converge with age on the count scale and parallel lines on the log scale (where points show the log of non-zero counts), so the group gap shrinks on the count scale and stays constant on the log scale. Whether the pattern is an interaction depends on the scale on which effects are assumed to add.



An interaction is a difference between differences, so the question is whether the group

difference itself changes with age. Since the gap narrows as the children grow older, the natural conclusion is that age and group interact. The data in Figure 1, however, were generated without any interaction. Errors follow a Poisson process in which age and group add on the log scale, so the two groups share the same age slope on that scale. A Poisson model with a log link recovers this structure, and on the log scale its fitted lines are parallel (Figure 1, Panel B). Mapped back onto the count scale, the same model produces two curves that converge with age, because a constant difference on the log scale is a constant ratio on the count scale, and the absolute gap between the groups therefore narrows as errors decline (Figure 1, Panel C). Whether an interaction is present thus depends on the scale on which effects are assumed to add. Additivity on the log scale implies non-parallel curves on the count scale, and additivity on the count scale would imply non-parallel curves on the log scale.

This ambiguity extends far beyond this single illustration, as interaction hypotheses are widespread in psychological research and often represent the central target of inquiry (Domingue et al., 2024; McCabe et al., 2022; Rohrer & Arslan, 2021). Whenever researchers ask whether an effect is moderated, or whether it holds only under certain conditions, they are asserting that the impact of one variable depends on another.

In psychology, this scale dependence has a characteristic source in measurement. Researchers routinely conceptualize constructs such as cognitive ability or symptom severity as continuous latent dimensions (Embretson & Reise, 2000; Wagenmakers et al., 2012). Those dimensions are rarely observed directly; what reaches the data are bounded, discrete quantities, such as counts, ratings, and accuracy scores that carry floors, ceilings, and chance levels. Each measurement scale imposes its own mapping from the latent dimension to the observed response. Where the mapping compresses, a constant effect on the latent dimension appears as a shrinking effect on the observed response: a group or condition closer to a bound has less room to move, and can look less affected by a manipulation, than a group or condition farther away from bounds (e.g., Figure 1).

Generalized linear models (GLMs) make this mapping explicit and give it a standard

vocabulary (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972), separating two choices that are easy to conflate, (1) the outcome family, which describes the response, and (2) the link function, which sets the scale on which effects combine (Box 1).

Box 1: Generalized linear models: outcome family and link function

The *outcome family* is the distribution assumed for the response. It defines which values the outcome can take and how observations vary around their expectation. The *link function*, $g(\cdot)$, maps the expected outcome, $\mu_i = E(Y_i)$, onto the linear predictor, η_i, the scale on which the model is additive. With two predictors and their interaction term,

$$g(\mu_i) = \eta_i, \qquad \eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

The inverse link, $g^{-1}(\cdot)$, maps this additive structure back onto the expected response. Its familiar role is to keep expected values within the admissible range of a constrained outcome: $\exp(\eta)$ keeps expected counts positive, and the inverse logit keeps probabilities between 0 and 1. Less obviously, this also fixes what the absence of an interaction, $\beta_3 = 0$, means in the fitted model. The product-term coefficient β_3 is therefore an interaction on the link scale. A response-scale interaction is a different quantity: a difference between expected differences on the observed outcome, for example, whether the group difference in expected errors changes with age in Figure 1. Unless the link is the identity, the two quantities can disagree (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Each family has a canonical link, which is the one that arises directly from the family's mathematical form, but it is a convenient default and other links are legitimate. A Gaussian linear model is the special case that combines a Gaussian family with an identity link, so the additive scale and the response scale coincide and the scale question is invisible. The model behind Figure 1 is a Poisson family with a log link. The family respects that errors are non-negative counts, and the link places additivity on the log scale.

Often the outcome family is chosen with care while the link function is left at its software

default, fixing the scale of additivity without deliberation. Two models can share a family and still place additivity on different scales. For example, a Gamma family suits response times; within that family, a log link makes effects additive on the log of the mean response time, a proportional change, and the canonical inverse link makes them additive in reciprocal means, a change in speed. A binomial family fits correct-versus-incorrect responses and leaves the link open among logit, probit, chance-corrected, and other forms.

The scale dependence of interactions has long been recognized. Loftus (1978) showed that interactions involving response probabilities are meaningful only relative to a mapping between observed and theoretical scales; Cox (1984) formalized interaction as a model-relative concept tied to transformation and interpretation. Busemeyer and Jones (1983) carried the same point into the measurement model, where the conclusion about how two variables combine depends on the model assumed to link responses to the quantities they measure. Lubinski and Humphreys (1990) showed that the specification choices alone can produce the appearance of an interaction where the generating process contains none.

More recently, Wagenmakers et al. (2012) separated removable from nonremovable interactions. A removable interaction can be undone by a monotonic transformation of the dependent variable, whereas a nonremovable one cannot. In this regard, J. C. Dunn and Anderson (2026) developed a removability test for linear models, while Rohrer and Arslan (2021) showed that the same interaction can be positive, null, or negative depending on the scale chosen for the outcome, so the appropriate test follows from the scale on which the hypothesis is posed. Rimpler and colleagues (2026) added that a product term assumes one predictor's effect changes at a constant rate across levels of the other, and so misrepresents any dependence that does not take that form, such as a threshold effect. Because the product-term coefficient of a GLM refers to the link scale (Box 1), a question about the response scale is answered with marginal effects (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026).

Domingue and colleagues (2024) demonstrated that outcome-model misspecification can produce false interaction findings for binary, count, and censored outcomes, and Liddell and

Kruschke (2018) showed that treating ordinal responses as metric can inflate error rates for interactions. Czado and Santner, and later Yu and Huang, studied the inferential consequences of link misspecification in binary GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019).

Despite the above cited literature spanning decades, to our knowledge no study has attempted to quantify how often applied psychological research makes explicit the outcome family and link that define the no-interaction baseline. The present article approaches that gap from two directions. We first document current practice with a preregistered review of recent empirical articles in five prominent psychology journals, quantifying how often interactions are tested, how often the outcome family and link function are made explicit, and which response formats predominate. We then develop a framework that separates outcome family, link function, and target metric, and work it through three cases, forced-choice accuracy with a chance floor, the choice between logit and probit links, and sum scores. Simulations quantify how often an inadequate link turns a purely additive structure into a statistically significant interaction, and whether standard diagnostics detect the misspecification.

Current Practice in Psychological Interaction Testing: A Preregistered Review

The review asks how often psychological articles test interactions, and how often they make the outcome family and link function explicit enough for a reader to know which interaction claim was tested. The coding is descriptive, so it documents reporting practice rather than the correctness of any substantive conclusion.

Review Method

The protocol was preregistered before any record was screened (https://osf.io/bdu73/), together with the coding data dictionary. The essentials are given here; the full protocol, sampling frame, and reliability are in Supplement A.

Following the preregistration, we sampled one journal from each of five areas: *Psychological Science* (general), *Journal of Experimental Psychology: General* (experimental), *Journal of Personality and Social Psychology* (social), *Developmental Science* (developmental), and *Psychological Medicine* (clinical). The five were selected on preregistered discretionary

criteria: a first-quartile 2023 Impact Factor, a broad rather than narrowly specialized scope, not primarily publishing reviews, and strong reputation in their field, as jointly judged by the coauthors. From each journal's full set of 2024 articles in Scopus, records were screened in a fixed random order until 40 eligible articles testing at least one interaction on an observed outcome were reached, yielding 200 coded articles. The preregistered stopping rule referred to interaction-testing articles in general; the restriction to observed outcomes was introduced during implementation. Because screening stopped at a fixed number of coded articles, the number of records screened, and therefore the denominator of the eligibility proportion, varied across journals. Eligible articles were empirical reports of original research; reviews, meta-analyses, editorials, theoretical and qualitative-only articles, replication-only studies, and psychometric validation studies were excluded. The use of latent-variable or structural-equation models was not a reason for exclusion. Articles whose interaction tests all concerned latent outcomes did not count toward the 40 per journal, because our family and link coding was defined for observed outcomes. Note that latent-variable models do not remove the scale problem per se, but move part of it to the measurement model and the definition of the latent scale. We treat this restriction as a deviation from the protocol and retain the resulting sample rather than reconstructing the point at which screening would otherwise have stopped. It concerned 8 of the 208 eligible interaction-testing articles; articles combining latent-variable analyses with at least one observed-outcome interaction were coded.

Each of the coded articles was independently coded by two researchers, with disagreements resolved by discussion or by consulting the other coauthors. Generative AI tools were used only as an optional aid to locate passages within an article; all coding used in the review is human.

For each article, the coders recorded five things: whether at least one tested interaction used a non-identity link; whether the link function was explicit; whether at least one interaction applied a Gaussian-identity analysis to a constrained outcome; whether at least one such interaction was statistically significant, or interpreted analogously in a Bayesian analysis; and

which response-variable types were analyzed.

By a Gaussian-identity analysis on a constrained outcome we mean a Gaussian linear regression with an identity link applied to an outcome with formal constraints, such as bounds, discreteness, positivity, or a known chance level. This is a conservative screening flag, applied only when the constraint was identifiable from the report and does not imply the substantive conclusion is wrong. The label is defined relative to the constrained process that generates the outcome (if the observed score were explicitly justified as the target metric, identity-link additivity would instead be a defensible modeling choice, a distinction we develop in the following sections).

Interrater agreement on the double-coded records was high (Cohen's κ between 0.66 and 0.81). Supplement A, Section S2, reports the preregistered variable definitions, the operational coding rules including borderline cases, per-variable agreement and coder margins, and the by-journal results.

Review Findings

First, interaction testing was common. Of the 367 articles screened, 345 met the eligibility criteria, of which 208 tested at least one interaction (60.3%). Under the stopping rule this proportion describes the records screened rather than a fixed-denominator sample, so we report it descriptively and treat the per-journal proportions as the more informative summary: these ranged from 39.4% in Psychological Medicine to 85.1% in Journal of Experimental Psychology: General. Of the interaction-testing articles, 8 (3.8%) tested interactions only on latent outcomes, and the remaining 200 formed the coded sample described below.

Second, model-scale reporting was uncommon. Among the 200 coded articles, 58 used at least one non-identity link function (29%), and only 46 made the link function explicit (23%). The link function was therefore not explicit in 77% of coded articles.

Even the 46 cases rest on a rule looser than the preregistered one, which required a declaration of the link and treated naming a family or a generalized linear model as insufficient. In coding we also accepted link-specific model names, because in practice they identify the link: "logistic regression" names the logit and "probit regression" the probit. We also accepted

"Poisson regression", which names a family whose canonical link is the log. Conversely, ANOVA, ordinary regression, and linear mixed models were not counted as explicit, although they imply the identity link, because there the identity scale is the default rather than a stated choice. The stricter preregistered rule would give a lower count.

Third, constrained outcomes were often analyzed on the identity scale. Among the 200 coded articles, 144 included at least one Gaussian-identity analysis on a constrained observed outcome (72%). Of these 144 cases, 124 reported at least one statistically significant interaction (86%). Because significance was coded only within the flagged analyses, this proportion describes those articles alone, and across all coded articles 62% combined a constrained outcome, an identity-link analysis, and a significant interaction claim. This is the same concern raised by work on outcome-model mismatch in interaction testing (Domingue et al., 2024; Liddell & Kruschke, 2018). Table 1 summarizes these results.

The pattern was present across the sampled journals, where the proportion of coded articles with a Gaussian-identity analysis on a constrained observed outcome ranged from 40% in Psychological Medicine to 95% in Psychological Science (Supplement A).

The outcomes used in interaction analyses were heterogeneous. The most common broad, non-exclusive categories (Table 2) were binary, accuracy, or proportion outcomes (84 articles), sum scores or composites (66 articles), ordinal, Likert, or rating outcomes (36 articles), response times or durations (30 articles), neural or physiological measures (23 articles), difference or distance scores (17 articles), counts or error counts (9 articles), and correlations or associations (8 articles). These broad categories combine the more detailed labels used to classify the response variables during coding. The operational definitions and decision rules used to distinguish them are reported in Supplement A.

What the Present Review Shows, What It Does Not

In these journals, then, interaction tests are common and the scale on which no interaction is assumed is rarely stated. The link function was explicit in only 23% of coded articles, and 72% applied an identity-link analysis to a constrained outcome. Bounded outcomes also dominated,

Table 1

Descriptive summary of the preregistered review. Gaussian-identity analyses on constrained outcomes are cases where the reported outcome had formal constraints that make identity-link additivity inappropriate as a generative model unless explicitly justified as an approximation or as the target observed-score metric. The interval on the eligibility row is reported for completeness only: screening stopped at 40 coded articles per journal, denominator was not fixed by design. Coded articles are eligible articles testing at least one interaction on an observed outcome.

Quantity	n	Denom.	%	95% CI (%)
Articles meeting eligibility criteria	345	367	94.0%	[91.1, 96.0]
Eligible empirical articles testing at least one interaction	208	345	60.3%	[55.0, 65.3]
Interaction-testing articles with at least one observed-outcome interaction (coded sample)	200	208	96.2%	[92.6, 98.0]
Interaction-testing articles with interactions on latent outcomes only	8	208	3.8%	[2.0, 7.4]
Coded articles using at least one non-identity link	58	200	29.0%	[23.2, 35.6]
Coded articles with explicit link function	46	200	23.0%	[17.7, 29.3]
Coded articles with Gaussian-identity analyses on constrained observed outcomes	144	200	72.0%	[65.4, 77.8]
Gaussian-identity cases with at least one significant interaction	124	144	86.1%	[79.5, 90.8]
Eligible empirical articles not testing interactions	137	345	39.7%	[34.7, 45.0]

Table 2

Broad response-variable categories in the coded articles. Categories are non-exclusive, so one article can contribute to more than one row.

Outcome type	n	Denom.	%
Binary / accuracy / proportions	84	200	42.0%
Sum scores / composites	66	200	33.0%
Ordinal / Likert / ratings	36	200	18.0%
Response times / durations	30	200	15.0%
Neural / physiological measures	23	200	11.5%
Difference / distance scores	17	200	8.5%
Counts / error counts	9	200	4.5%
Correlations / associations	8	200	4.0%

with binary, accuracy, or proportion outcomes, sum scores, and ordinal ratings together appearing in 152 of the 200 coded articles (76%). For approximately continuous outcomes, additivity on the observed scale can be a defensible target, whereas for bounded and discrete outcomes it is a stronger assumption, and treating them as metric distorts estimates, error rates, and interaction conclusions (Bürkner & Vuorre, 2019; Liddell & Kruschke, 2018).

A descriptive review cannot establish how often published interactions are scale artifacts, because that judgment requires knowing the scale on which the data-generating process is additive. J. C. Dunn and Anderson (2026) provide a direct test of removability, but it requires cell-level summaries and error estimates that the reviewed articles often did not report, and many designs used mixed-effects models, whose output does not reduce to cell-level means. The remainder of the paper therefore turns to simulations, where the generating process is known by construction, and focuses on the binomial and aggregated responses that dominated the corpus, including sums of binary or ordinal items.

The Link Function Problem in Three Common Cases

To show what link misspecification does to interaction tests, we conducted three simulation studies. Each asks whether a data-generating process that is purely additive on one scale yields a statistically significant interaction when it is modeled on another scale.

1. *Forced-Choice Accuracy*: This case holds the binomial outcome family fixed and evaluates whether the link function must explicitly encode a task-implied chance floor (e.g., a .50 guessing threshold).
2. *Within-Family Link Choice (Logit vs. Probit)*: This case holds both the binary family and the 0-to-1 bounds fixed, testing whether choosing between these two standard link functions alters the results.
3. *Sum Scores*: This case holds the latent generating process fixed and evaluates whether the bounded total shows an interaction that the latent scale does not.

All three take up the case in which the outcome family is already broadly plausible and the link alone sets a disputable no-interaction baseline, which is the common situation for bounded and chance-limited psychological outcomes. Work on misspecification has largely varied the outcome model as a whole (Domingue et al., 2024; Liddell & Kruschke, 2018), whereas here the link carries most of the disagreement about what counts as no interaction.

The simulations share one design. Data are generated with no product term on the generating scale, and models assuming additivity on that scale and on plausible alternative scales are fitted to the same data. The rate at which the matched model rejects a null product term is the *nominal rejection rate*. A product term estimated under a mismatched model is a genuine property of that model's scale (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022), correct where it is computed and misleading only when the claim is carried to the generating scale. Because that scale is known in a simulation, we call such a result a *pseudo-interaction*, and the rate at which it reaches significance the *pseudo-interaction detection rate*.

The scenarios were chosen to represent plausible psychological data. Each reproduces a common interaction-testing situation, in which two well-established and large main effects, for

example a group difference combined with a condition effect or a developmental trend, raise the question of whether one effect moderates the other. Most scenarios set these effects strong enough to push expected values close to a floor, ceiling, or chance level while keeping them inside the observed range, which is where link curvature acts most strongly, and scenarios away from the bounds serve as a comparison. Technical details are reported in Supplement A, and Supplement B reports a broader sensitivity atlas that varies sample size, main-effect magnitude, the intercept, and selected design parameters. A final simulation, in the next section, examines whether standard diagnostics identify the misspecified link.

Forced-Choice Accuracy and the Non-Zero Chance Floor

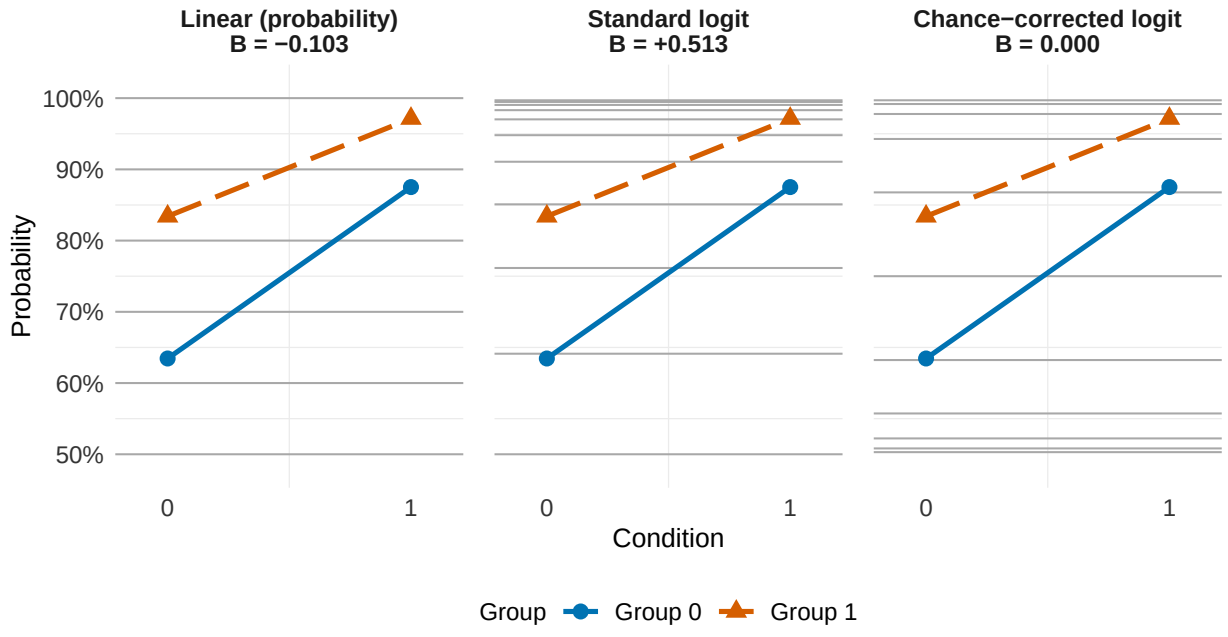
If a participant answers Y_i of k trials correctly, a binomial sampling model, $Y_i \sim \text{Binomial}(k, p_i)$, is the natural starting point. The family requires the expected accuracy to lie between 0 and 1. When guessing is a plausible process for the task, the range that matters is narrower, since a participant who knows nothing still gets a share c of the trials right, $c = .50$ in a two-alternative or true/false task and $c = 1/m$ with m alternatives, so expected accuracy lies between c and 1, even though observed proportions can fall below c through sampling variation. Accuracy data of this kind are analyzed in several ways in practice, on the raw proportion scale with a Gaussian identity model, with a standard binomial logit or probit, and, less often, with a link that encodes the chance floor.

These specifications differ in which of the two ranges they impose. The Gaussian identity model imposes neither, since it ignores the trial process, assumes additivity in percentage-correct units, and can predict outside the 0-to-1 range. The other two keep the binomial family and differ only in the link. The standard logit or probit, $p_i = F(\eta_i)$ with F the logistic or standard-normal CDF, holds expected accuracy between 0 and 1 and places the lower asymptote at 0, whereas the chance-corrected link maps the linear predictor onto the narrower interval from c to 1, $p_i = c + (1 - c)F(\eta_i)$, which for $c = .50$ becomes $p_i = .50 + .50\,F(\eta_i)$. An interaction term in the standard model therefore tests additivity over the full 0-to-1 range, whereas the same term in the chance-corrected model tests additivity on a performance-above-chance scale, so the same four

cell probabilities can imply different interactions on the three scales (Figure 2).

Figure 2

A minimal 2×2 *design crosses a condition (x-axis) with a group (color), shown on three scales. The four cell probabilities are identical in every panel; only the scale on which the interaction is evaluated changes. The data were generated with no condition-by-group interaction on the chance-corrected logit scale, which has a* .50 *chance floor, so the interaction coefficient is exactly zero on that scale. The same cells imply a positive interaction on the standard logit scale and a negative interaction on the raw probability scale. Light horizontal lines are equally spaced on each panel's own link scale, so equal steps on that scale map to unequal steps on the probability axis.*



The chance-corrected link is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and below-chance performance is not substantively meaningful for the target claim. In the simplest case, the chance level can be fixed by design, for example $c = .50$ in a two-alternative task. In other cases, analysts may estimate the lower asymptote or report sensitivity analyses over plausible values of c. Lapses, response biases, item heterogeneity, changes across trials, or systematic below-chance performance may instead

require richer psychometric solutions such as item-response models (e.g., Embretson & Reise, 2000; Knoblauch & Maloney, 2012; Wichmann & Hill, 2001). For example, the 3-parameter logistic Item Response Theory model estimates a guessing parameter for each item, where the chance-corrected link fixes a single chance level for the whole design (Embretson & Reise, 2000). The chance-corrected link is thus a design-level approximation, and item-level measurement models remain the appropriate tool when the data and the scientific question call for that additional structure.

Simulation 1: Design and Results

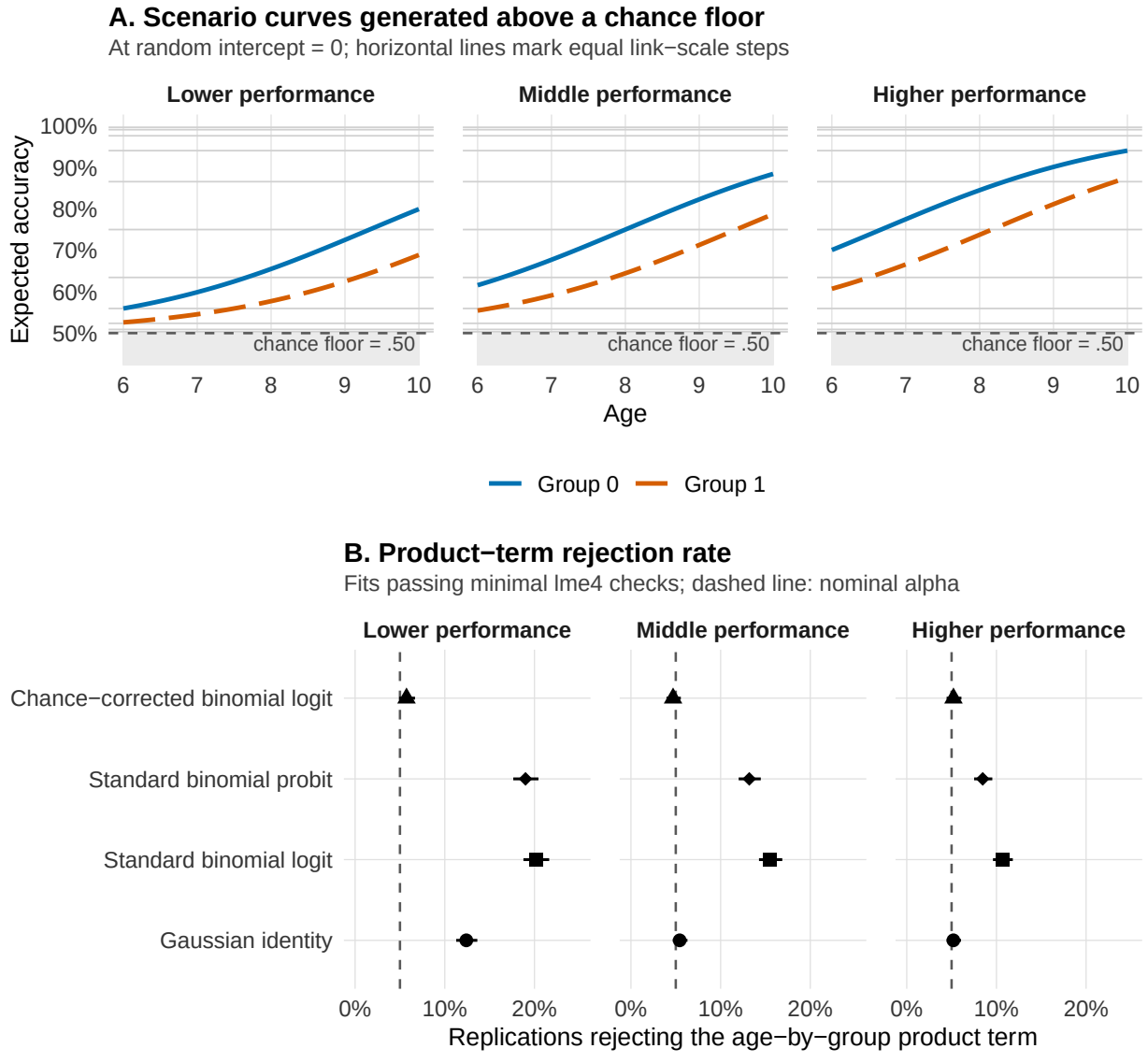
In each of 3000 replications we generated 20 trial-level Bernoulli responses for each of $N = 250$ participants. Age and group were constant across a participant's trials. The data-generating model included a subject random intercept corresponding to a latent-scale ICC of .30. The probability of a correct response followed a chance-corrected logit with a .50 floor, a positive age effect, a group difference, and no age-by-group product term on the conditional chance-corrected-logit scale. Three scenarios differ only in overall performance, set by shifting the fixed intercept. The deterministic curves in Panel A are evaluated at random intercept 0.

Each simulated dataset was analyzed with four mixed models sharing the same subject random intercept: a Gaussian identity LMM applied to the binary trial outcome, standard binomial-logit and binomial-probit GLMMs, and the matched chance-corrected binomial-logit GLMM. The matched model was fitted with `lme4::glmer` and `psyphy::mafc.logit(2)`, setting `nAGQ = 10` (default 1), which evaluates the likelihood more accurately at some computational cost, because with the default up to nearly half of these fits were flagged (Supplement A). Each model received one fit attempt. We report rejection rates separately for fits passing basic convergence checks and for fits flagged by numerical problems, because the latter can yield unreliable Wald tests. Exact criteria are given in Supplement A.

For the matched chance-corrected model, the primary rate is the nominal rejection rate among numerically acceptable fits. For the three alternative links it is the pseudo-interaction detection rate among acceptable fits. Rates among flagged fits are descriptive apparent rejection

Figure 3

Simulation 1: forced-choice accuracy with a chance floor and a subject random intercept. The product term is zero on the conditional chance-corrected-logit scale. Panel A shows fixed-effect curves at random intercept 0 across three performance levels; the grey region lies below the .50 chance floor. Panel B shows the primary product-term rejection rates among fits passing the minimal fit-quality checks. The dashed line marks $\alpha = .05$.



rates, not valid inferential error rates.

The matched chance-corrected model rejected the product term in 5% to 6% of acceptable fits. The alternative links produced pseudo-interactions, most often near the chance floor. In the lower-performance scenario the standard logit detected an interaction in 20% of replications, the standard probit in 19%, and the Gaussian identity in 12%. In the higher-performance scenario these rates fell to 11%, 8%, and 5%.

The matched model was also the hardest to fit. Even with `nAGQ = 10`, from 7% to 12% of its fits were flagged, against at most 4% for the alternatives. Flagged fits are not a minor nuisance here, because they rejected the product term in 75% to 78% of cases. An analyst who ignored the warnings would therefore obtain a pseudo-interaction rate of 10% to 14% from the model that is correct by construction, comparable to some of the wrong links. The chance-corrected link is the ideal solution in principle, but in a GLMM it requires attention to fitting options and convergence warnings.

Within-Family Link Choice

In the forced-choice case the standard link was wrong for the task by construction, because it placed the lower asymptote at 0 when the design fixed it at chance. The link-function problem can also arise in less obvious cases. Logit and probit are both standard for binary data, respect the same 0-to-1 bounds, and usually produce very similar fitted probabilities. They differ mainly in interpretation, since logit coefficients are log-odds, whereas probit coefficients are expressed on a standard-normal latent scale, which makes probit the usual choice when modeling psychological quantities such as ability, signal strength, and response tendency (P. K. Dunn & Smyth, 2018). The choice between them is therefore often guided by convention, interpretation, or software, with little expected impact on substantive conclusions (Agresti, 2002; Yu & Huang, 2019). Yet even two nearly interchangeable links define slightly different scales of additivity. Equal differences under one link are not exactly equal under the other, especially toward the tails, and this alone can alter the estimated product term.

Simulation 2: Design and Results

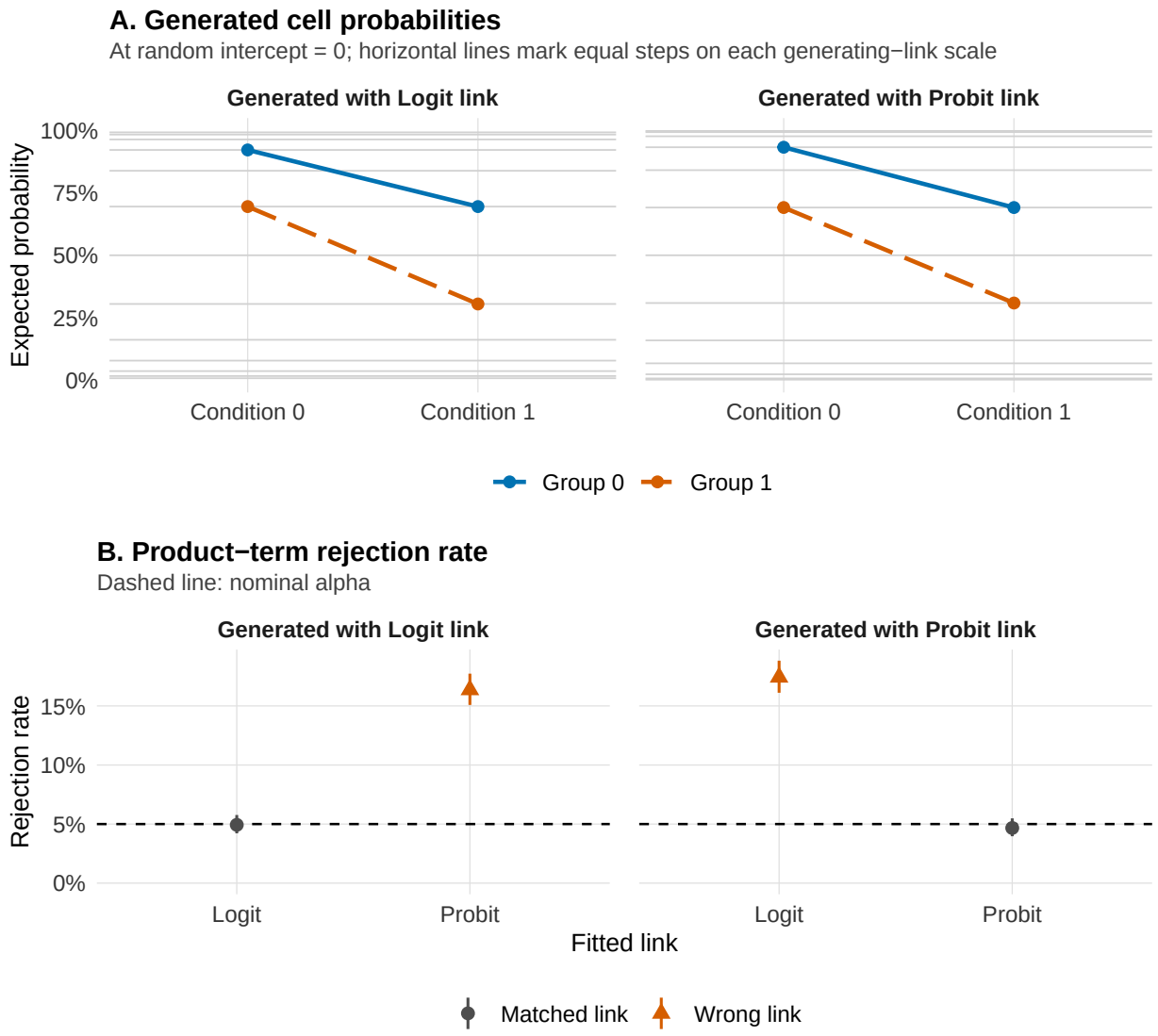
In each of 3000 replications we generated repeated binary responses for $N = 700$ participants in a two-group, two-condition design, with 15 trials per participant-condition cell and a subject random intercept (latent intraclass correlation .30). The large sample was chosen so that the small systematic discrepancy between logit and probit links would be detectable, and it is larger than in a typical psychological study. The condition-by-group interaction term was zero on the generating link scale. We crossed the generating link with the fitted link, using logit and probit in both roles, and fitted every model as a random-intercept GLMM, recording how often each returned a significant condition-by-group interaction term at $\alpha = .05$. Figure 4 shows the results; the full crossed-link rates are tabulated in Supplement A.

When the fitted link matches the generating link, the nominal rejection rate stays near .05, at 5% for logit fitted to logit-generated data and 5% for probit fitted to probit-generated data. Crossing the links raises the pseudo-interaction detection rate to 17% when a logit model is fitted to probit-generated data and 16% when a probit model is fitted to logit-generated data, roughly three times the nominal rate. The wrong-link detection rates are smaller than in the forced-choice case, and they appear among links that share the same family, the same 0-to-1 bounds, and nearly overlapping fitted curves in the middle of the probability range, the setting many applied analysts treat as interchangeable.

These rates reflect a fixed quantity. Under the wrong link the same cell probabilities imply a small but deterministic nonzero product term on the fitted scale. Re-expressing the cell probabilities of each generating model on the other link scale gives a product term of -0.112 for logit-generated data fitted with a probit and 0.216 for probit-generated data fitted with a logit, against exactly zero whenever the links match (Supplement A, Section S5). These values hold at random intercept 0; the median GLMM estimates are smaller (-0.041 and 0.078), because changing the link also changes how the random intercept enters the model, but they remain a fixed, nonzero property of the fitted scale. Supplement B varies the intercept, which shifts where the four cell probabilities sit between 0 and 1, and shows that detectability changes

Figure 4

Simulation 2: logit versus probit in repeated-trial random-intercept GLMMs. Data are generated with no condition-by-group product term on the generating link scale. Panel A shows the corresponding cell probabilities under logit and probit links. Panel B shows product-term rejection rates for each generating–fitted link combination. Matched links remain near nominal alpha, whereas wrong links produce pseudo-interactions. Error bars are Wilson 95% confidence intervals; the dashed line marks $\alpha = .05$.



non-monotonically with it, so the same link mismatch is nearly undetectable at some probability levels and much easier to detect at others.

Sum Scores

Many psychological outcomes are computed as total sum scores across questionnaire items, task items, symptoms, errors, or ratings. In the two cases above the outcome was a directly observed response, and only the link was in question. A sum score is different, because the score itself is already the result of an analytic choice. What is observed are the individual item responses, which are binary or ordinal; the total is a summary of them, produced by a scoring rule that gives every item the same weight and every point of the scale the same width. In psychological research the construct of interest is often the underlying, generally continuous dimension, so the total can be seen as a remapped version of that continuum. The link question then arrives together with a prior question about which metric the claim is about.

The bounds matter because the construct may have no floor or ceiling, whereas the total score has both. The relation between the two can be described by how much change on the latent dimension is needed to raise the total by one point, and this amount is not constant. In the middle of the score range a small latent change moves the total by a point, because many items are close to changing category. Near either bound a much larger latent change is needed, because most items are already at their lowest or highest category. A constant latent effect therefore translates into fewer raw-score points as the total approaches the floor or the ceiling, and the same holds wherever the instrument is less informative along the latent dimension, so the effect is not limited to the endpoints. A Gaussian identity model fitted to the total assumes the opposite, namely that a predictor shifts the score by the same number of points at every level of the other predictor. When one group lies closer to a bound, a predictor with the same latent effect in both groups moves that group's score by fewer points, and the model reports the difference as an interaction.

The problem we raise sits next to a recent, wider debate about whether sum scores should be replaced by factor scores or latent-variable models. Some argue that sum scores are transparent, reliable enough for many purposes, comparable across studies, and tied to how instruments are

scored in applied settings (Widaman & Revelle, 2023); the counterargument is that a total encodes strong and usually untested assumptions about how items represent the construct (McNeish, 2023; McNeish & Wolf, 2020). Without entering that debate, we add a narrower point that it leaves aside. A model of the manifest total defines the absence of an interaction in raw-score points, so an interaction found there is an interaction on the score, and it becomes a claim about the construct only under the additional assumption that the two are linearly related.

When item-level data are available, the measurement model itself can instead be modeled explicitly (Bürkner & Vuorre, 2019). Often, however, researchers deliberately work with the total score for the practical reasons listed above. One option is then a family that respects the bounds and the discreteness of the total, for example a beta-binomial likelihood for a bounded count (Wilcox, 1981). Here we use a deliberately unusual alternative that changes only the link, keeping the observed total under a Gaussian likelihood, rescaling it to the proportion of the maximum score, and applying a probit link, as in Simulation 3 below. This is an *ad hoc* device outside standard practice, and it illustrates whether the interaction conclusion changes if additivity is placed on a nonlinear scale that compresses the expected values toward the score bounds. It remains a model of the total, it will not recover the construct, and the Gaussian likelihood is still misspecified in its variance, since a bounded score has a variance that depends on its mean, so the probit link changes only the scale on which additivity is defined and leaves the distributional form as it is. Read against the sum-score debate, it is a precaution for researchers who have decided to analyze totals anyway and who accept that the interaction still needs a scale on which additivity is plausible.

Simulation 3: Design and Results

In each of 3000 replications, we generated data for $N = 600$ participants from a continuous latent variable with a continuous predictor x, a group effect, and no x-by-group interaction. The latent values generated 9 ordinal items scored 0 to 3, which were summed to a total ranging from 0 to 27. Item thresholds were shifted across three scenarios so that expected totals fell in the lower, middle, or upper part of the score range.

We analyzed each total in three ways: a Gaussian identity model of the observed sum score, a Gaussian-probit bounded-score model, and the true latent generating scale. The bounded-score model applies a probit link to a continuity-corrected score proportion and rescales predictions to the original sum-score metric. We used probit because its standard-normal linear predictor provides a simple correspondence with the latent scale. We recorded the proportion of significant x-by-group interactions at $\alpha = .05$. Results are shown in Figure 5, with per-scenario rates in Supplement A, Section S4.

The latent benchmark remained close to the nominal .05 rejection rate in every scenario, from 4% to 5%. The identity model rejected in 72% of replications in the lower range, 8% in the middle range, and 36% in the upper range, so it came closest to the nominal rate in the middle, where the sum score is approximately linear in the latent variable, and its rate rose as expected totals approached a bound. Compression of the latent scale makes the same latent group difference appear to change across levels of the predictor. The bounded-score probit model rejected in 8%, 3%, and 4% of replications in the same three ranges. Its predicted means respect the score bounds, although it is still a model of the manifest total.

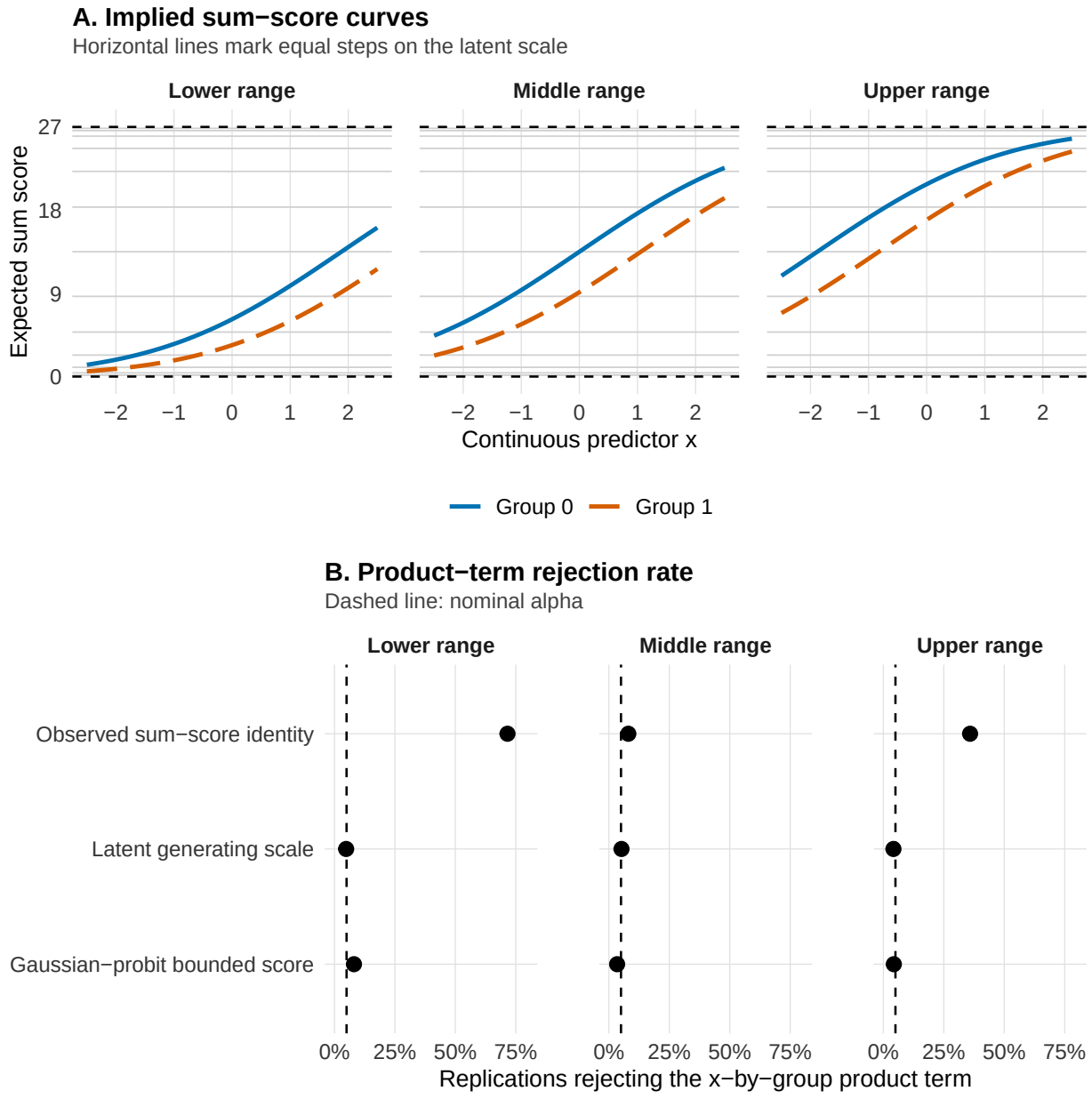
This pattern mainly depends on where observations fall within the score range. Across the broader scenarios in Supplement B, the position of expected scores within the range affected identity-model rejection much more than sample size, item count, or latent dispersion. Raw-score additivity can approximate latent-scale additivity when scores remain away from the bounds. The bounded-score probit model can therefore serve as a sensitivity analysis. For claims about interactions between latent constructs, an explicit measurement model provides a more direct approach, for example SEM, item-level ordinal models, or IRT-inspired models. These approaches still depend on assumptions about thresholds, links, loadings, and latent-response scales, but they make those assumptions explicit.

Can Standard Diagnostics Flag a Wrong Link?

The simulations above assume that the analyst accepts the fitted model. In practice, a researcher who finds an interaction may check the model before reporting it. We examined three

Figure 5

Simulation 3: sum scores as bounded and discrete outcomes. Data are generated with no x-by-group product term on the latent scale. Panel A shows how the same additive latent structure maps onto sum scores in the lower, middle, and upper parts of the score range. Panel B shows product-term rejection rates for the latent benchmark and two observed-score models. The latent model remains near nominal alpha, whereas the observed-score models can produce pseudo-interactions. The dashed line marks $\alpha = .05$. The latent benchmark is available only in simulation.



checks based on different strategies. A Pregibon-style link test probes the assumed link, residual diagnostics screen for misfit of any kind, and AIC compares candidate models.

The Pregibon test refits the model adding the squared fitted linear predictor, $\hat{\eta}^2$, as a predictor (Pregibon, 1980). If the link is adequate, this term adds no information. A significant coefficient indicates curvature that the link does not explain. For residual diagnostics, we used the simulated quantile residuals of DHARMa (Hartig, 2024; see also P. K. Dunn & Smyth, 1996). Under an adequate model they are uniform between 0 and 1, whatever the family. Misfit appears as non-uniformity, over- or underdispersion, or patterns over fitted values and predictors. AIC compares only the models it is given. Our candidate models share the response, the family, and the formula, and differ only in the link. They therefore have the same number of parameters, and AIC ranks them by log-likelihood alone.

None of the above checks is specific to the link. A significant Pregibon term targets the link, but it also flags any omitted nonlinearity or interaction. Moreover, the product term absorbs part of the curvature produced by a wrong link. With a continuous predictor, some curvature remains within groups and can in principle be detected. In a saturated 2×2 design no curvature remains in the fixed effects, because any link can reproduce the four cell means.

The monotonic linear model of J. C. Dunn and Anderson (2026) tests scale dependence more directly, asking whether additivity can be recovered under any monotonic transformation. All pseudo-interactions simulated here are removable in this sense. However, the test requires cell-level data from a factorial design and has no straightforward extension to mixed models, whereas most of our scenarios include continuous predictors, trial-level random effects, or both.

Diagnostic Simulation

The five scenarios are based on examples introduced earlier; sample sizes and coefficients are reported in Supplement A. Each scenario keeps the outcome family and the interaction formula fixed and changes only the link. The fitted model is therefore wrong about the scale on which effects add, and about nothing else. *Count errors* follows the Poisson-log example of Figure 1 and fits it with an identity link. *Chance-floor accuracy* takes the lower-performance

scenario of Simulation 1, with trial-level responses and a subject random intercept, and compares standard-logit and chance-corrected-logit GLMMs with the same random effects. *Binary repeated trials* follows the probit-generated, logit-fitted case of Simulation 2 with 300 participants instead of 700. *Binary continuous predictor* repeats that case with a continuous predictor instead of the two-level condition, so that residual patterns can be examined along a predictor. *Gamma mean response time* generates response times from a Gamma model with a log link and fits the inverse link.

In each replication we applied the checks to the misspecified interaction model, because this is the model an analyst would inspect after finding a significant interaction. The residual checks covered uniformity, dispersion, quantile patterns over fitted values, and patterns over the focal predictor. The last was assessed with a quantile check for a continuous predictor and with a comparison across design cells in the 2×2 case. All residual checks were run with the package defaults (DHARMa 0.5.0). AIC compared the misspecified model with the target model. We also applied the residual checks and the Pregibon test to the correctly specified model fitted to the same data, to see how often each flags a model that is right. A check is informative about the link only if it flags the misspecified model more often than the correct one, so Table 3 reports the two rates side by side. A systematic evaluation of residual diagnostics for link misspecification is beyond our scope. Rates are based on the replications that returned the quantities required by each diagnostic; denominators are reported in Supplement A.

Table 3 shows how the checks differ in sensitivity. The residual checks flagged the wrong link in up to 91% of replications in the count scenario, and in at most 7% in the other four. The Pregibon test flagged it in 99% of count replications, 32% of Gamma replications, and 12% of chance-floor replications. In the binary scenario with a continuous predictor it flagged 5%, about its rate under the correct link. It cannot be computed in the saturated 2×2 design. AIC preferred the target link in 80% to 99% of replications.

The count scenario is the only one in which the wrong link distorted the fit enough for the Pregibon test and the most sensitive residual check to flag it almost always. In the other four the

Table 3

Diagnostic simulation across five link-misspecification scenarios.

Scenario	Link: correct (generating) → wrong (fitted)	Pseudo-interaction rate, wrong link	DHARMa flagging rate, correct / wrong link	Pregibon flagging rate, correct / wrong link	AIC prefers correct link
Count errors	log → identity	0.97	0.00–0.05 / 0.01–0.91	0.05 / 0.99	0.99
Chance-floor accuracy	chance-corrected logit → standard logit	0.20	0.00–0.08 / 0.00–0.04	0.14 / 0.12	0.98
Binary repeated trials	probit → logit	0.10	0.00–0.08 / 0.00–0.07	N.I.	0.96
Binary continuous predictor	probit → logit	0.07	0.00–0.04 / 0.00–0.03	0.05 / 0.05	0.96
Gamma mean response time	log → inverse	0.31	0.00–0.02 / 0.00–0.06	0.04 / 0.32	0.80

Note. In each replication the same data were fitted with the correct (generating) link and with a wrong link; family and formula were the same. Pseudo-interaction rate: proportion of replications in which the wrong-link model detected an interaction absent on the correct link scale. Correct / wrong link: flagging rate for the correct-link model / for the wrong-link model; a check is informative only if the second rate clearly exceeds the first. DHARMa: range across the four residual checks, run with package defaults. Pregibon: added-term link check; N.I. = not identifiable in the saturated 2×2 design, where the squared linear predictor is collinear with the fixed effects. AIC prefers correct link: proportion of replications in which the correct-link model had the lower AIC; the two models have the same number of parameters, so indistinguishable links would give about .50. Denominators and confidence intervals are reported in Supplement A.

residual checks almost never flagged the misspecification, and AIC gave the most consistent signal, but only because the target link was among the candidates. The chance-floor scenario is the most instructive. Pseudo-interactions occurred in 20% of replications, yet the residual checks flagged the misspecified standard logit in at most 4%. Applied to the correct chance-corrected model, they flagged it in up to 8%, and the Pregibon test in 14%, slightly more often than the misspecified model. The checks thus did not favor the right model. In all other scenarios no check flagged more than 8.4% of correctly specified models. In the two logit-probit scenarios pseudo-interactions were less frequent, at 10% and 7%. Supplement A reports the full results.

From Link Choice to Sensitivity Analysis

In practice, an interaction analysis should state the three choices it depends on: the outcome family, the link that defines additivity, and the metric the claim refers to. It should also test whether its conclusion holds under plausible alternatives (Table 4). Sometimes the observed metric is the relevant target. For example, a two-point difference on a clinical or educational score may be what matters in practice, and additivity in raw-score units may then be reasonable. The point is that this is a substantive choice, which should be made deliberately and reported. When possible, the link should be justified by theory, task structure, or measurement.

When the link can affect the interaction conclusion, we suggest a sensitivity analysis across plausible links as the main robustness check. The removability test of J. C. Dunn and Anderson (2026) addresses the same problem directly, testing any monotonic transformation of the outcome. However, besides having no straightforward extension to mixed models, it does not decide which transformations are admissible. The family, the bounds, the chance level, and the meaning of the outcome strongly restrict the plausible mappings. We suggest testing the interaction under each defensible link, reporting the corresponding response-scale contrast, and stating whether the conclusion is stable or depends on the link. If it depends on the link, this should be disclosed (McCabe et al., 2022; Mize, 2019; Mustillo et al., 2018; Rohrer & Arel-Bundock, 2026).

Table 4

Practical starting points for link-aware interaction testing across common psychological outcomes.

Outcome type	Plausible starting point	Minimal sensitivity check
Forced-choice accuracy	Binomial logit/probit; chance-corrected link when the task floor is part of the estimand.	Compare standard logit/probit and chance-corrected links; report convergence and response-scale contrasts.
Binary / proportion outcome	Binomial logit/probit; consider cloglog for asymmetric processes.	Compare logit/probit; inspect calibration and contrasts near bounds or rare-event regions.
Likert / ordinal item	Ordinal logit/probit; metric model only with a clear observed-scale target.	Compare ordinal and metric summaries; report category probabilities or response-scale contrasts.
Sum score / composite	Gaussian identity when many levels stay far from bounds; otherwise consider item-level, ordinal, bounded-count, or bounded-score models.	Inspect bounds, skew, and score concentration; compare with item-level or bounded-score sensitivity models when feasible.
Counts / error counts	Poisson or negative binomial, usually with log link; identity only with an observed-count target.	Check overdispersion and zero inflation; compare count contrasts across plausible count models.
Response time / skewed positive outcome	Lognormal, Gamma log/inverse, Inverse Gaussian, or transformed Gaussian, matched to absolute, proportional, reciprocal, or transformed change.	Compare raw-, log-, and response-scale summaries; inspect tail fit and predicted values.

Discussion

The same data can support different interaction conclusions under different plausible specifications, which is the familiar scale dependence of interactions (Cox, 1984; J. C. Dunn & Anderson, 2026; Loftus, 1978; Wagenmakers et al., 2012). In a GLM that dependence is carried by two choices, the outcome family and the link function. Our review found interaction tests to be common in all five journals sampled, the link function usually left implicit, and identity-link analyses applied to many outcomes for which observed-scale additivity is not self-evident.

The simulations then showed what that can cost. With two main effects present, a link that misplaces additivity turned a purely additive generating process into a significant product term even when the outcome family was correct. In forced-choice accuracy the binomial family is correct and, in the simulated process, the error lies entirely in a canonical link that places the lower asymptote at 0 rather than at the chance level. The matched chance-corrected model, however, was the hardest to fit, and its flagged fits produced spurious interactions in most cases, so the ideal model requires care to be a usable one. The logit-probit comparison showed that the problem survives when the response bounds are correct as well, between two links that look alike over most of the range. In the sum-score case the choice extends beyond the link to the outcome itself. A total is a scoring rule applied to item responses, so an interaction estimated on it refers to raw-score points, and adopting a family that respects the bounds and the discreteness of the total still leaves open whether additivity in points stands for additivity in the construct.

This difficulty is of a different kind from the one documented in work on the precision of interaction estimates (Baranger et al., 2023; Castillo et al., 2026). There the obstacle is noise, and more data helps. Here the product term induced by the wrong link is a fixed property of that scale, so a larger sample estimates it more precisely and reports it as significant more often.

The diagnostic simulation asks whether ordinary post-fit checks flag link misspecification. Residual checks, Pregibon-style tests, and AIC answer different questions, and AIC can only rank the links it is given. Outside the count scenario, residual checks rarely detected the wrong link, and the Pregibon test did so in at most a third of replications. In the chance-floor case they even

flagged the correct model more often than the wrong one, so a clean diagnostic report did not mean that the scale was right. More generally, these diagnostics answer a question about fit within a supplied candidate set, which is different from whether an interaction claim holds on the scale the researcher means (P. K. Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

Three limitations bound the scope of these conclusions. The review was descriptive and covered five prominent journals across major areas of psychology in a single year, so its estimates describe reporting practice in such outlets and leave the field as a whole unmeasured, and a broader mapping across journals, years, and subfields would be needed to support reporting guidance. The review also coded observable reporting features and did not assess removability in each article. The simulations, finally, are demonstrations of a mechanism. They target a single regime, in which two established main effects push expected values into the region where link curvature is strongest, and they say nothing about how often that regime arises or about the further complications of real data.

To conclude, interaction claims rest on a scale that is rarely stated. Naming it separates three questions that a software default otherwise settles at once, which family suits the response, which link defines additivity, and which metric the claim is about. The robustness question then takes a definite form, whether the conclusion survives the links that remain defensible. A result that holds across them is robust to that choice. One that does not is still a result, and it should be reported as conditional on the scale on which it was estimated.

Author Contributions

Author roles were classified using the Contributor Roles Taxonomy (CRediT; https://credit.niso.org/) as follows — Conceptualization: E. Toffalini, F. Gambarota, T. Feraco; Investigation: M. Calderan, L. Sità, E. Toffalini, F. Gambarota, T. Feraco; Formal Analysis: L. Sità, E. Toffalini, F. Gambarota; Methodology: E. Toffalini, F. Gambarota; Software: E. Toffalini, F. Gambarota; Writing - Original Draft: M. Calderan, E. Toffalini; Writing - Review & Editing: F. Gambarota, T. Feraco, and L. Sità; Visualization: E. Toffalini. All authors approved the final version of the manuscript and agree to be accountable for the work.

Conflicts of Interest

The authors declare that there were no conflicts of interest with respect to the authorship or the publication of this article.

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Data, Code and Materials

All data, code, and supplementary files are accessible at https://doi.org/10.5281/zenodo.XXXXXXX

Preregistration

The protocol of the literature review is available at https://osf.io/bdu73/.

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